

Report
Of
Frontend Web Developer

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, Ashutosh Shukla, do hereby declare that the internship report titled “Frontend Web Development” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: ASHUTOSH SHUKLA

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to YuvaIntern for giving me the opportunity to undergo this internship as a Frontend Web Developer Intern. I am thankful to Mr. Kounal Gupta, Founder, YuvaIntern, and the entire team for their guidance and support throughout the duration of this internship program.

I am also grateful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing continuous academic support and encouragement, and for the opportunity to undertake this internship as part of my B.Tech CSE curriculum.

I am thankful for the aspiring guidance, invaluable constructive criticism and friendly advice extended to me during this work. I am sincerely grateful for the truthful and illuminating views shared with me on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: ASHUTOSH SHUKLA

Student Name: ASHUTOSH SHUKLA

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INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **Ashutosh Shukla** successfully completed an internship with **YuvaIntern** in the role of **Virtual Data Exploration Intern**.

Throughout the tenure of this internship, Ashutosh demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend Ashutosh's hard work and wish them continued success in all future endeavors.

Date of Issue: August 26, 2026

FOR AND ON BEHALF OF YUVAINTERN

A handwritten signature in black ink, appearing to read "Kounal".



Kounal Gupta

Founder, [YuvaIntern.com](https://yuvaintern.com)

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Figure A.1: Certificate of Experience issued by YuvaIntern to ASHUTOSH SHUKLA.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during my Summer Internship undertaken with YuvaIntern, in the role of Frontend Web Developer Intern. The internship was conducted remotely over a duration of five weeks, from July 22, 2026 to August 26, 2026, as part of the requirement of the B.Tech Computer Science and Engineering programme at IILM University, Greater Noida.

The internship focused on the practical aspects of frontend web development, beginning with the conversion of a wireframe into a static webpage and progressing through responsive design, JavaScript-based interactivity, accessibility and performance optimization, and finally the simulation of a Single Page Application (SPA) using client-side routing. Each week of the internship built upon the concepts covered in the previous week, resulting in a structured and cumulative learning experience.

This report documents the objectives, technologies, methodology and outcomes of the internship on a week-by-week basis, along with the official documents supporting the completion of the programme.

4.2 Organization Profile

YuvaIntern is identified in the internship offer letter and certificate of experience as the organization with which this internship was undertaken. YuvaIntern operates in association with NSDC (National Skill Development Corporation), under the initiative “Re-Imagine Future”, and offers virtual internship programs and job simulations to student interns. The internship program was led by Mr. Kounal Gupta, Founder, YuvaIntern.com.

As documented in the offer letter, the internship was conducted on a remote basis and involved working on tasks and virtual job simulations related to frontend web development, with the intern expected to comply with the organization's policies and confidentiality requirements. The materials provided do not include further corporate or organizational history; accordingly, this profile is limited to the information available in the official offer letter and certificate.

4.3 Problem Statement

Modern websites are accessed across a wide variety of devices, ranging from desktop monitors to tablets and mobile phones, each with different screen sizes and interaction methods. A webpage designed without responsiveness, interactivity and accessibility in mind fails to deliver a consistent and usable experience across this range of devices and users.

The practical problem addressed during this internship was the development of a webpage – starting from a given wireframe – into a fully responsive, interactive and accessible frontend interface, and subsequently extending it into a simulated Single Page Application with working client-side navigation. This required the application of core web technologies (HTML5, CSS3 and JavaScript) along with responsive design techniques, accessibility standards and performance testing tools.

4.4 Project Objectives

- To study and analyze a given wireframe and convert it into a static webpage using HTML5 and CSS3.
- To implement responsive web design using Flexbox, CSS Grid and media queries for desktop, tablet and mobile devices.
- To integrate JavaScript for interactive user experience, including DOM manipulation, event handling and form validation.
- To enhance webpage performance and accessibility using semantic HTML, ARIA attributes, keyboard navigation and Lighthouse testing.
- To develop a simulation of a Single Page Application (SPA) using dynamic content loading and the Browser History API.
- To gain hands-on, practical experience in frontend web development through weekly, task-based project work.

4.5 Scope of the Project

The scope of this internship covered the complete frontend development workflow, from converting a static wireframe into a webpage to building a simulated multi-page application experience within a single HTML document. It included:

- Week 1: Conversion of a wireframe into a responsive static webpage using HTML5 and CSS3.
- Week 2: Implementation of responsive web design using Flexbox, CSS Grid, media queries and responsive navigation.
- Week 3: Integration of JavaScript for interactivity, including a mobile navigation menu and contact form validation.
- Week 4: Enhancement of webpage performance and accessibility through semantic HTML, ARIA attributes and Lighthouse testing.
- Week 5: Development of a Single Page Application simulation with client-side routing using the History API.

The scope of the internship was limited to frontend technologies (HTML5, CSS3 and JavaScript) and did not involve backend development, databases, or external frameworks or libraries beyond what is listed in this report.

4.6 Technologies and Tools Used

Table 4.6 lists the technologies and tools used during the course of the internship. Only the technologies actually applied in the weekly tasks are listed below; no additional frameworks or libraries were used.

Technology / Tool	Purpose
HTML5	Semantic structure and markup of webpages
CSS3	Styling, layout design and visual presentation
Flexbox	Flexible one-dimensional layout design
CSS Grid	Two-dimensional, grid-based page layout
Media Queries	Responsive breakpoints for desktop, tablet and mobile
JavaScript (DOM)	Interactivity, dynamic content and event handling
Event Listeners / event.preventDefault()	Handling user interactions and form submission behaviour

Technology / Tool	Purpose
ARIA Attributes	Improving accessibility for assistive technologies
Browser History API (pushState, popstate)	Client-side routing for the SPA simulation
Chrome Developer Tools	Responsive design testing and debugging
Lighthouse	Performance and accessibility auditing

Table 4.6: Technologies and tools used during the internship.

4.7 System Architecture

The frontend development work carried out during the internship followed a progressive workflow, where each stage built upon the output of the previous stage, moving from static structure to a fully interactive, accessible and navigable interface.

Stage	Focus Area	Primary Activity
1	Wireframe Analysis	Study the given wireframe and plan the page structure
2	Static Page Development	Build page structure with HTML5 and style with CSS3
3	Responsive Layout	Apply Flexbox, CSS Grid and media queries for multi-device support
4	Interactivity	Add JavaScript, DOM manipulation, event handling and form validation
5	Accessibility & Performance	Apply semantic HTML, ARIA attributes and Lighthouse testing
6	SPA Simulation	Implement client-side routing using the Browser History API

Table 4.7 / Figure 4.7.1: Wireframe → Structure & Style → Responsive Layout → Interactivity → Accessibility & Performance → SPA Simulation.

4.8 Methodology

The internship followed a structured, week-by-week methodology in which each week introduced a specific frontend development concept and built directly upon the work completed in the previous week. The methodology followed for each week is described below.

4.8.1 Week 1: Wireframe to Responsive Static Web Page

Objective: To study a given wireframe and convert it into a static webpage using HTML5 and CSS3.

Description: The assigned wireframe was studied and analyzed to understand the intended page structure and layout. The webpage was then built by writing semantic HTML5 markup for the page structure and CSS3 for styling, using Flexbox and CSS Grid to arrange page elements and media queries to adapt the layout for different screen sizes.

Technologies Used: HTML5, CSS3, Flexbox, CSS Grid, Media Queries.

Implementation: The wireframe layout was translated into HTML elements, and CSS Flexbox and Grid were used to position the header, content sections and footer. Media queries were added to adjust the layout at different breakpoints.

Testing: The webpage was tested by resizing the browser window and using device emulation to check its appearance on desktop, tablet and mobile screen sizes.

Challenges & Problem-Solving: Aligning elements consistently across different screen sizes required adjusting the Flexbox and Grid properties and refining the media query breakpoints through repeated testing.

Learning Outcome: This week helped build a strong foundation in translating a design/wireframe into working HTML and CSS code, and in structuring a webpage using modern CSS layout techniques

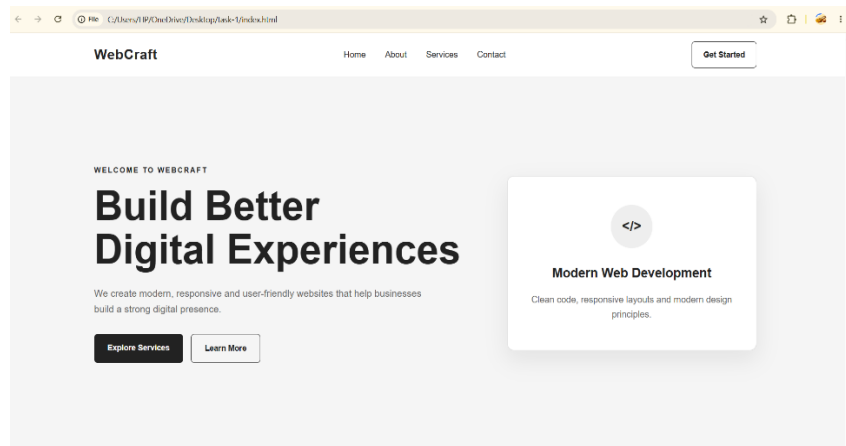


Figure 4.8.1: Responsive static webpage developed in Week 1 (Desktop and Mobile view).

4.8.2 Week 2: Implementation of Responsive Web Design

Objective: To further develop the responsive static webpage with a focus on responsive navigation, images and cross-device consistency.

Description: Building on the Week 1 webpage, additional responsive design techniques were implemented, including a responsive navigation bar and responsive images/elements that adapt to different screen widths. The layout was refined using Flexbox and CSS Grid in combination with media queries.

Technologies Used: HTML5, CSS3, Flexbox, CSS Grid, Media Queries, Responsive Navigation, Chrome Developer Tools.

Implementation: A responsive navigation structure was implemented, and images and content elements were made to resize and reflow appropriately across breakpoints. Chrome Developer Tools were used throughout to preview the page at different device widths.

Testing: The webpage was tested across simulated desktop, tablet and mobile viewports using Chrome Developer Tools to verify that the navigation and layout behaved correctly at each breakpoint.

Challenges & Problem-Solving: Ensuring the navigation bar remained usable on smaller screens required rethinking its layout for mobile widths, which was resolved through additional CSS rules within the media queries.

Learning Outcome: This week strengthened practical skills in building fully responsive interfaces and in using browser developer tools for responsive design testing and debugging.

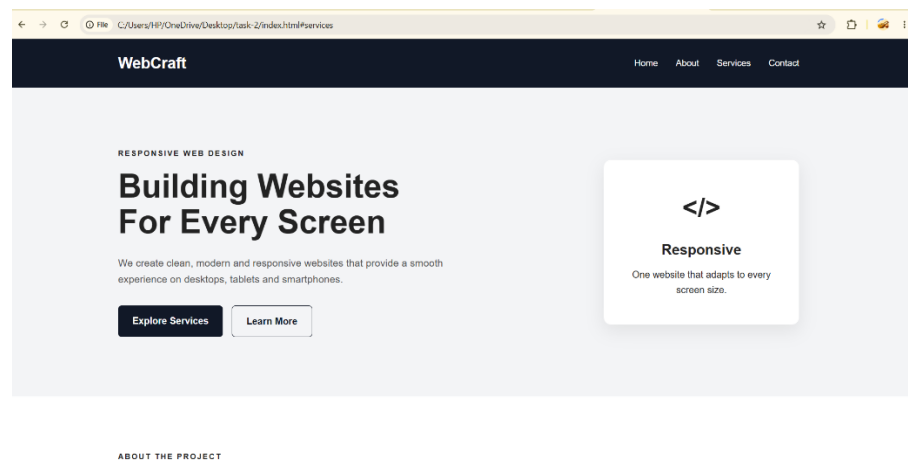


Figure 4.8.2: Responsive navigation and layout implementation in Week 2.

4.8.3 Week 3: Integrating JavaScript for Interactive User Experience

Objective: To add interactivity to the webpage using JavaScript, including a mobile navigation menu and a validated contact form.

Description: JavaScript was integrated into the webpage to introduce interactivity. This included DOM manipulation, event listeners for user actions, a functional mobile navigation menu, a dynamic message/button interaction, and a contact form with client-side validation, including email format validation and handling of successful form submission.

Technologies Used: JavaScript, DOM Manipulation, Event Listeners, `event.preventDefault()`.

Implementation: Event listeners were attached to the mobile menu toggle and form elements. The contact form was validated on submission, checking required fields and email format, and `event.preventDefault()` was used to prevent default form submission behaviour while validation and feedback messages were handled through JavaScript.

Testing: The mobile navigation menu and form validation logic were tested manually by submitting valid and invalid form inputs and verifying that appropriate feedback and behaviour occurred in each case.

Challenges & Problem-Solving: Handling form validation edge cases (such as invalid email formats and empty fields) required writing and testing multiple validation conditions before the form behaved reliably.

Learning Outcome: This week provided practical experience in DOM manipulation, event-driven programming and client-side form validation using vanilla JavaScript.

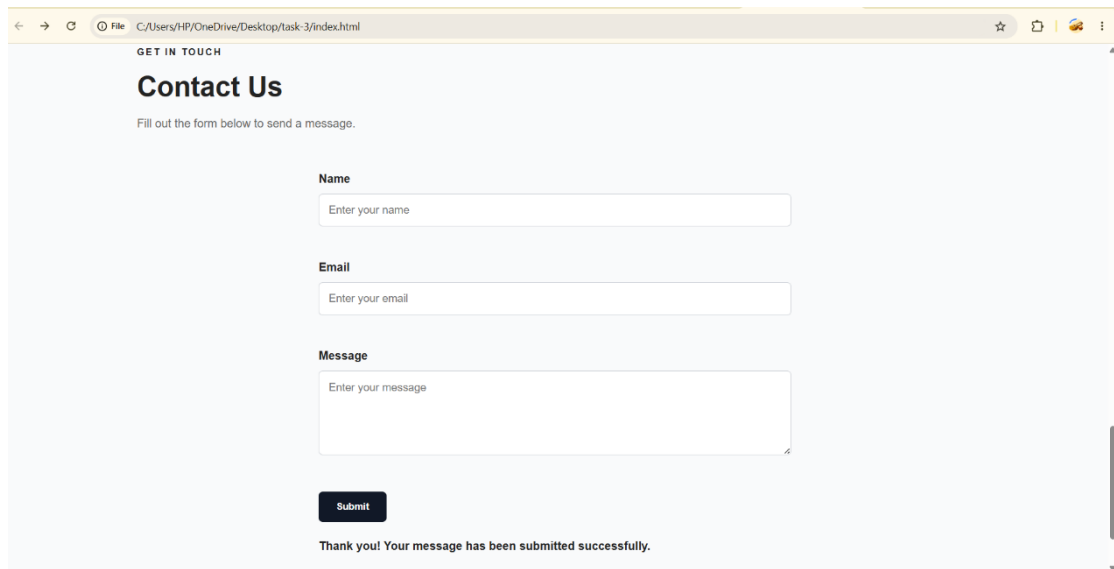
A screenshot of a web browser window displaying a contact form. The browser's address bar shows the file path 'C:/Users/HP/OneDrive/Desktop/task-3/index.html'. The page has a light blue header with the text 'GET IN TOUCH' and a main heading 'Contact Us'. Below the heading is a subtext 'Fill out the form below to send a message.' The form consists of three input fields: 'Name' with a placeholder 'Enter your name', 'Email' with a placeholder 'Enter your email', and 'Message' with a placeholder 'Enter your message'. A dark blue 'Submit' button is located below the message field. At the bottom of the form, a message reads 'Thank you! Your message has been submitted successfully.'

Figure 4.8.3: JavaScript-based mobile menu and contact form validation in Week 3.

4.8.4 Week 4: Enhancing Web Page Performance and Accessibility

Objective: To improve the accessibility and performance of the webpage using semantic HTML, ARIA attributes and testing tools.

Description: The existing webpage was reviewed and enhanced for accessibility and performance. This involved using semantic HTML elements, adding a skip-to-content link, ensuring keyboard navigability, adding visible focus indicators, applying appropriate ARIA attributes to interactive elements, organizing CSS for a lightweight implementation, and supporting reduced motion preferences.

Technologies Used: Semantic HTML, ARIA Attributes, CSS, Lighthouse.

Implementation: Accessibility improvements such as a skip-to-content link, visible focus states and ARIA labels were added to interactive elements and navigation. CSS was reorganized to keep the implementation lightweight, and a reduced-motion media query was added for users with motion sensitivity preferences.

Testing: The webpage was tested for keyboard navigation (tab order and focus visibility) and audited using Lighthouse for accessibility and performance scoring.

Challenges & Problem-Solving: Identifying all elements that required ARIA attributes and ensuring a logical keyboard tab order took several rounds of testing and adjustment before the Lighthouse accessibility score improved satisfactorily.

Learning Outcome: This week built an understanding of web accessibility principles (WCAG-aligned practices) and of using Lighthouse as a tool to audit and improve accessibility and performance.

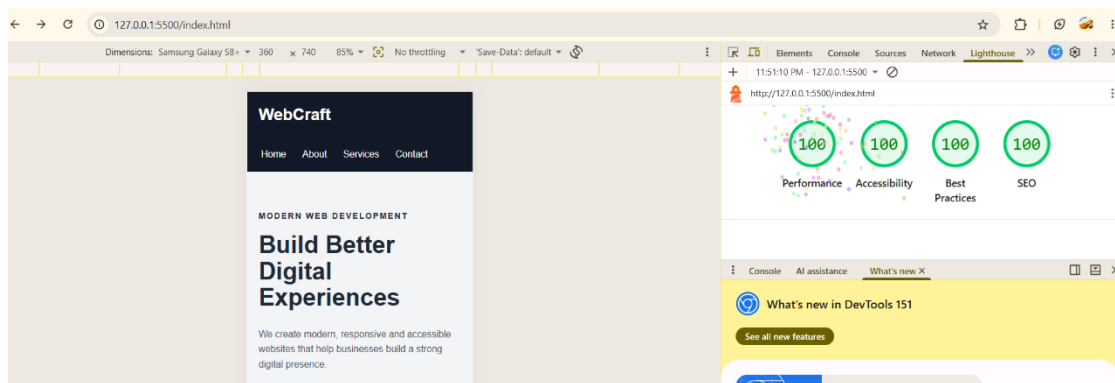


Figure 4.8.4: Lighthouse accessibility and performance audit results from Week 4.

4.8.5 Week 5: Developing a Single Page Application (SPA) Simulation

Objective: To simulate a Single Page Application experience with client-side routing, without full page reloads.

Description: A Single Page Application simulation was developed using HTML5, CSS3 and JavaScript. Dynamic content loading was implemented through DOM manipulation, and client-side routing was handled using the Browser History API (`history.pushState()` and the `popstate` event) to switch between Home, About, Services and Contact sections while supporting the browser's Back and Forward navigation.

Technologies Used: HTML5, CSS3, JavaScript, Browser History API (`pushState`, `popstate`).

Implementation: Separate content sections for Home, About, Services and Contact were created and shown or hidden dynamically through JavaScript based on the active route. `history.pushState()` was used to update the URL and browser history without

reloading the page, and the popstate event was used to handle Back/Forward navigation correctly.

Testing: The SPA simulation was tested by navigating between routes using the on-page links as well as the browser's Back and Forward buttons, and by verifying that the URL and displayed content remained in sync across different screen sizes.

Challenges & Problem-Solving: Keeping the browser history state synchronized with the displayed content – particularly when using the Back and Forward buttons – required careful handling of the popstate event and was resolved through iterative testing and debugging.

Learning Outcome: This week introduced the core concepts behind client-side routing and Single Page Applications, and how the Browser History API can be used to simulate multi-page navigation within a single HTML document.

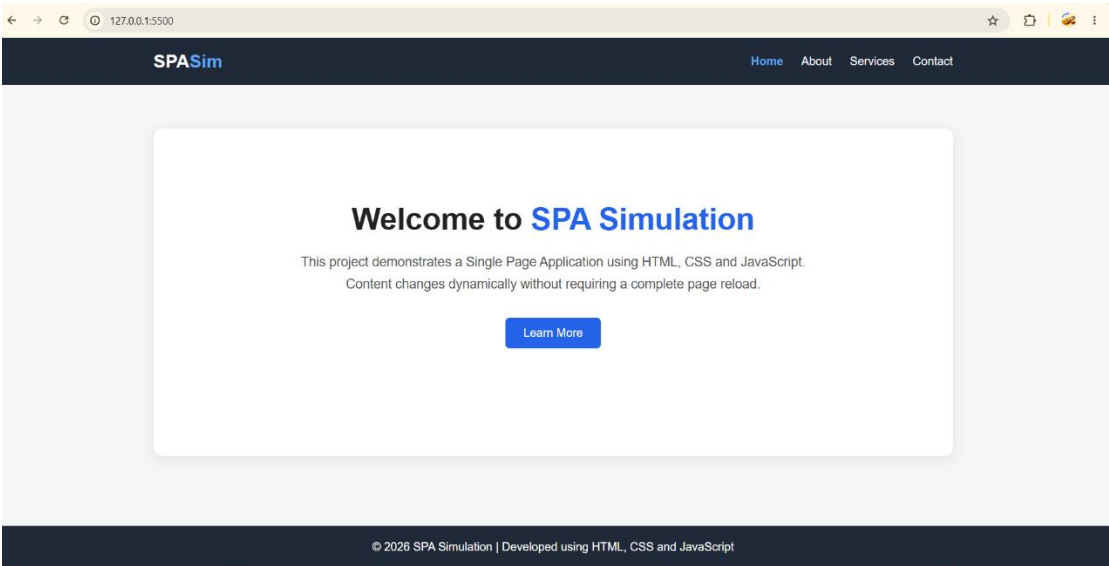


Figure 4.8.5: Single Page Application simulation with client-side routing in Week 5.

Table 4.8: Week-wise summary of internship tasks.

Week	Task Title	Key Focus
1	Wireframe to Responsive Static Web Page	HTML5, CSS3, Flexbox, Grid, Media Queries
2	Implementation of Responsive Web Design	Responsive navigation, images, cross-device testing
3	Integrating JavaScript for Interactive UX	DOM manipulation, events, form validation
4	Enhancing Performance and Accessibility	Semantic HTML, ARIA, Lighthouse testing
5	Single Page Application (SPA) Simulation	Client-side routing, History API

4.9 Expected Outcomes

Through the course of this five-week internship, the following outcomes and capabilities were developed:

- Ability to convert a wireframe into a working, responsive webpage using HTML5 and CSS3.
- Practical proficiency in responsive design using Flexbox, CSS Grid and media queries.
- Experience in adding interactivity to webpages using JavaScript, DOM manipulation and event handling.
- Ability to build and validate web forms on the client side, including email validation and submission handling.
- Understanding of web accessibility principles and practical experience implementing semantic HTML and ARIA attributes.
- Familiarity with using Chrome DevTools and Lighthouse for responsive testing, debugging and accessibility/performance auditing.
- Understanding of the core concepts of Single Page Applications and client-side routing using the Browser History API.
- Overall, a structured, hands-on foundation in frontend web development suitable for further academic and professional application.

4.10 Certificates of Completion and Communication Proof

The internship completion certificate issued by YuvaIntern is presented in Chapter 3 of this report (Figure A.1). The official internship offer letter, which documents the internship role, duration and terms, is presented below as further supporting proof of the internship.



Figure A.2: Internship Offer Letter issued by YuvaIntern to ASHUTOSH SHUKLA.

5. BIBLIOGRAPHY / REFERENCES

This report has been prepared using the official internship documents, the official report format supplied by IILM University, and standard frontend web development documentation. The following references represent the source materials used for this report:

- YuvaIntern Internship Offer Letter, issued to Ashutosh Shukla, Frontend Web Developer Intern, dated July 22, 2026.
- YuvaIntern Certificate of Experience, issued to Ashutosh Shukla, dated August 26, 2026.
- Internship Report Format, School of Computer Science and Engineering, IILM University, Greater Noida, U.P.
- Mozilla Developer Network (MDN Web Docs) – HTML, CSS and JavaScript documentation, developer.mozilla.org.
- W3C Web Accessibility Initiative (WAI) – Accessibility guidelines, w3.org/WAI.
- Google Chrome Developers – Lighthouse documentation, developer.chrome.com/docs/lighthouse.